Runze Li

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Education

New York University — Brooklyn, NY, Unitied States
Master in Computer Science — *Sep 2023 – Exp. May 2025*

Tianjin University — Jinnan District, Tianjin, China
Bachelor in Computer Science and Technology | ***GPA: 3.66/4.0*** — *Sep 2019 – Jun 2023*

University of Oxford — Remote
Exchange Program of Artificial Intelligence — *Jan 2022 – Mar 2022*

Technical Skills

Programming : C, C++, Java, Python, LaTeX, HTML, CSS, JavaScript, Vue, SQL, Markdown
Frameworks : Node.js, SpringBoot, Mybatis, PyTorch, TensorFlow, NumPy, Pandas, Matplotlib
Dev Tools : Visual Studio Code, PyCharm, IntelliJ, Git, Linux, Postman

Experience

Front End Developer Intern — Apr 2022 – Jun 2022
Xiaomi Corporation — *Beijing, China*

- Designed and developed Web-based multimedia frame player.
- Built the module of **video player**.
- Created the module of **scene interaction of controller**, including speed control, pause, etc.
- Implemented the module of **FPS index monitoring**, which can monitor the number of video frame.
- Used: **HTML, CSS, JavaScript, Vue**.

Projects

Student Management System (Full-Stack) — Jun 2023 – Aug 2023
Personal Project – (GitHub)

- Built the frame by using **SpringBoot and MyBatis**.
- Used **MySQL** to store students' information.
- Tested stability of the system by using **Postman**.
- The system includes operations such as adding, deleting, modifying and querying, and paging query.
- Used: **Java, HTML, CSS, JavaScript, Vue**.

Dynamic Token Normalization in Vision Transformer — Mar 2023 – Jun 2023
Graduation Dissertation – (GitHub)

- Implemented a **dynamic token normalization (DTN)** in Vision Transformer using Python.
- Built a framework of Vision Transformer based on **PyTorch**.
- Designed a **dynamic learnable parameter** which can enable DTN to use LN and IN simultaneously in the same calculation formula.
- Improved Top-1 accuracy on CIFAR-10 by **1.08%** and CIFAR-100 by **2.46%**.
- Used **Matplotlib** to analyze the weight of LN and IN.

MiniMIPS32 Instruction Set Simulator — Sep 2021 – Nov 2021
Team Coursework: Semester project – (GitHub)

- Led a team of 6: organized meetings, assigned tasks, helped debug, etc. Final grade of **94%**.
- Implemented a instruction set emulator for **MIPS32 instruction set** using C based on Linux.
- Based on the MIPS32 instruction, the simulator can **perform 57 instructions**, including operation, memory access and branch jump, etc.
- The simulator supports single-step execution, breakpoints, display register information, etc.
- MiniMIPS32 can handle a variety of **synchronous exceptions**.